

Like and Comment Functionality - Implementation Details

Overview

This feature adds comprehensive like and comment functionality to community posts in The Writers Corner application.

Database Changes

New Models

1. Like Model

- Fields: id, userId, submissionId, createdAt
- Unique constraint on userId + submissionId (prevents duplicate likes)
- Index on submissionId for query performance
- Cascade delete when user or submission is deleted

2. Comment Model

- Fields: id, content, userId, submissionId, createdAt, updatedAt
- Indexes on submissionId and userId for query performance
- Cascade delete when user or submission is deleted

Model Updates

- **User:** Added relationships to likes and comments
- **ExerciseSubmission:** Added relationships to likes and comments

API Endpoints

Likes

- GET /api/community/posts/[id]/likes - Get like count and user's like status
- POST /api/community/posts/[id]/likes - Add a like
- DELETE /api/community/posts/[id]/likes - Remove a like

Comments

- GET /api/community/posts/[id]/comments - Get all comments for a post
- POST /api/community/posts/[id]/comments - Create a new comment
- PUT /api/community/posts/[postId]/comments/[commentId] - Edit a comment (owner only)
- DELETE /api/community/posts/[postId]/comments/[commentId] - Delete a comment (owner only)

Updated Endpoints

- GET /api/community/posts - Now includes like/comment counts and user's like status

Authentication

All endpoints require authentication via NextAuth. Anonymous users cannot:

- Like posts
- View like counts
- View comments
- Add, edit, or delete comments

Frontend Features

Like Button

- Click to like/unlike a post
- Visual feedback when liked (filled heart icon, rust color)
- Shows like count when > 0
- Real-time updates without page reload

Comment Section

- Expandable/collapsible comment section
- Comment form with textarea
- List of all comments in chronological order (newest first)
- Each comment shows:
 - Author name
 - Timestamp
 - “edited” indicator if modified
- Edit/delete buttons for comment owner

Comment Management

- **Add:** Users can add new comments via the comment form
- **Edit:** Comment owners can edit their own comments
- **Delete:** Comment owners can delete their own comments with confirmation
- Real-time updates to comment counts

UI/UX Considerations

- Follows existing design system (vintage typewriter theme)
- Uses Radix UI components and Tailwind CSS
- Animations via Framer Motion
- Responsive design
- Accessible components

Security

- All mutations require authentication
- Users can only edit/delete their own comments
- Input validation on comment content
- Proper error handling and user feedback

Performance

- Database indexes on frequently queried fields
- Lazy loading of comments (only fetched when section expanded)
- Optimistic UI updates where appropriate
- Efficient queries with proper includes

Migration Steps

To deploy this feature:

1. Run Prisma migration: `npx prisma migrate dev --name add_likes_and_comments`
2. Generate Prisma client: `npx prisma generate`
3. Restart the application

Testing Recommendations

1. Test like/unlike functionality
2. Test comment CRUD operations
3. Test authentication requirements
4. Test ownership restrictions on edit/delete
5. Test real-time count updates
6. Test UI responsiveness
7. Test with multiple users simultaneously